Table 1.

Body mass (g) and brain, neocortex, hippocampus, cerebellum, and medulla oblongata volume data (mm^3) for the 64 species considered in our study (brain mass given converted to volume: brain volume = brain mass/1.036, Baron *et al.*, 1996). We included the cerebellum and medulla oblongata as control/neutral regions for which we had no *a priori* reason to expect systematic differences in volume between migratory and sedentary species.

Family	Species	Body (g)	Brain (mm^3)	Neocortex (mm^3)	Hippocampus (mm^3)	Medulla Oblongata (mm^3)	Cerebellum (mm^3)	Category	References
Pteropodidae	*Cynopterus brachyotis*	33.9	946.9	287.6	80.9	68.3	122.7	Sedentary	9, 18
	Cynopterus horsfieldi	59.3	1325.3	432.3	110.3	92.7	180.5	Sedentary	18
	Eidolon helvum	262	4140.9	1406	258.1	273.7	463.3	Migratory	9, 18, 20, 23, 24, 29
	Epomophorus wahlbergi	74.1	1751					Migratory	18, 19
	Macroglossus minimus	14.6	541.5	155.7	47	41.3	79.2	Sedentary	9
	Myonycteris torquata	36.2	1116.8	336.5	85.7	82.2	155.8	Migratory	9, 18, 20, 29
	Pteropus alecto	595	6794.4	2716	357.4	392.1	843.7	Migratory	9, 20
	Pteropus poliocephalus	695	6978.8	2710	358.5	431.6	880.3	Migratory	7, 9, 20, 21, 30
	Pteropus scapulatus	375	5173.7	2078	267.8	284.3	682	Migratory	8, 9, 20, 21
	Syconycteris australis	14.7	550.2	170	48.4	37.9	64.9	Sedentary	9
Hipposideridae	*Hipposideros commersoni*	99.2	723	171.7	47.8	86.5	128.4	Migratory	6, 19
	Hipposideros lankadiva	48.5	707.5	154.5	48.8	93	127.3	Migratory	11
	Triaenops persicus	13.7	261.6	48.1	17.4	38.7	66.1	Migratory	19
Rhinolophuidae	*Rhinolophus ferrumequinum*	23.5	337.8					Sedentary	9, 12, 13, 26, 28
	Rhinolophus hipposideros	4.6	143.8	19	13.6	23	30.4	Sedentary	9, 12, 13, 26, 28
	Rhinolophus landeri	9.7	267.4	50.6	19.1	35.3	56.3	Migratory	19
	Rhinolophus mehelyi	9.7	299.2					Sedentary	9, 13
Megadermatidae	*Cardioderma cor*	26	646.7	186.3	32.3	71.5	111.7	Sedentary	9
	Lavia frons	23.4	621.6	171.4	25.8	67.7	138.8	Sedentary	9
	Macroderma gigas	119.8	1644.8	510.5	68.9	186.7	301.4	Sedentary	9
Rhinopomatidae	*Rhinopoma microphyllum*	29.4	373.6	77.5	25.8	51.3	60.4	Migratory	1, 11
Emballonuridae	*Coleura afra*	11.5	248.1	52.5	12.4	33.1	49.1	Sedentary	9
	Emballonura monticola	5.3	160.2	32.1	7.61	21	31.8	Migratory	17
	Emballonura semicaudata	7	177.6	36	10.1	25.7	36	Migratory	17
	Saccopteryx bilineata	8.4	223	43.9	12.2	27.2	37.8	Sedentary	9
	Taphozous australis	24	499	101.8	25.5	65.8	104.1	Sedentary	9
	Taphozous mauritianus	29	531.8	105.9	25.4	70.4	114	Migratory	19
	Taphozous melanopogon	22.7	517.4	113.2	23.7	67	112.7	Migratory	1, 11
Nycteridae	*Nycteris grandis*	29.8	684.4	187	36.2	80.3	139	Sedentary	9
	Nycteris thebaica	8.9	311.8	85.7	17.4	39.6	58	Sedentary	19
Noctilionidae	*Noctilio leporinus*	58	1141.9	361.7	50.5	124.4	194.1	Sedentary	9
Phyllostomidae	*Desmodus rotundus*	36.3	964.3	312.5	42.4	80	166.4	Sedentary	9
	Diaemus youngi	34.6	944					Sedentary	9
	Diphylla ecaudata	30.9	770.3	231.1	41	63.1	132	Sedentary	9
	Glossophaga soricina	9.9	379.3	88	31.4	37.9	81.6	Sedentary	10, 18
	Leptonycteris curasoae	24.5	588.8	140.3	47.7	62.2	119.4	Migratory	2, 9, 10, 15, 16, 20, 22
	Leptonycteris nivalis	22.8	565.6	144.3	42.2	56	104.4	Migratory	9, 10, 16, 18, 20
	Phyllostomus hastatus	91.8	1480.7	438.9	79.8	140.2	281.3	Sedentary	9
	Vampyrum spectrum	173	2497.1	757.7	110.4	233.5	532.7	Sedentary	9
Vespertilionidae	*Barbastella barbastellus*	5.8	176.6					Migratory	9, 12, 13
	Eptesicus fuscus	13.6	229.7	43.3	19.7	31.1	38.7	Sedentary	2, 3, 9, 18
	Eptesicus serotinus	23.4	299.2	55.7	27.6	47.1	55.2	Sedentary	3, 9, 28
	Ia io	53.3	734.6	141.3	47.1	95.3	145.6	Migratory	18
	Lasiurus borealis	7.8	164.1	27.9	9	23.4	40.9	Migratory	2, 3, 4, 9, 12, 18, 20, 22
	Myotis bechsteinii	9.5	255.8	36.3	25.5	39.4	42.5	Sedentary	3, 9, 13
	Myotis dasycneme	15	299.2	46.5	25.2	49.3	49.8	Migratory	3, 9, 12, 13, 26, 28
	Myotis daubentonii	7	222					Migratory	3, 9, 12, 13

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	Myotis lucifugus	7.9	163.1					Migratory	2, 3, 5, 8, 9, 12, 18, 20, 22
	Myotis myotis	25.6	468.1	78.3	35.6	70.3	87.1	Migratory	3, 9, 12, 13, 27
	Myotis nattereri	7	212.4	34.8	21.4	31.6	37.4	Sedentary	3, 9, 13, 28
	Nyctalus noctula	27	351.4	54.9	19.2	57.6	80.5	Migratory	3, 9, 12, 13, 18, 20, 26, 27, 28
	Perimyotis subflavus	5.3	120.7	16.6	10.9	18.3	21.3	Migratory	3, 9, 22
	Pipistrellus kuhlii	6.3	160.2					Sedentary	3, 9, 13
	Pipistrellus nanus	3.9	96.5	15.7	6.27	15.2	18.5	Sedentary	3, 19
	Plecotus auritus	6.6	229.7					Sedentary	3, 9, 13, 28
	Scotophilus dinganii	24.3	421.8	87.2	22.7	58.2	81.5	Sedentary	3, 6
	Scotophilus nigrita	101.3	1009.7					Migratory	19
Miniopteridae	*Miniopterus australis*	7.1	196.9	37.6	15	27.2	39.7	Migratory	17
	Miniopterus inflatus	14.9	313.7	55.9	19	43.8	79.5	Migratory	17
	Miniopterus schreibersii	11.4	245.2					Migratory	3, 9, 10, 12, 13, 18, 20, 25, 26, 27, 28
Molossidae	*Chaerephon pumila*	10.3	234.6	44.3	12.2	36.3	41.8	Sedentary	18
	Eumops perotis	47.5	774.1					Sedentary	18, 22
	Molossus ater	33.6	507.7	113.8	24.4	66	93.8	Sedentary	18
	Mops condylurus	27.2	441.1	102	22.9	58.6	74.4	Migratory	19
Migratory (n = 31, mean±se)		86.0 ± 30.3	1152.8 ± 339.2	435.3 ± 166.2	73.9 ± 21.7	100.6 ± 22.9	184.0 ± 49.9		
Sedentary (n = 33, mean±se)		31.3 ± 6.2	625.0 ± 91.0	186.1 ± 34.9	38.7 ± 5.5	68.1 ± 9.8	119.1 ± 21.1		

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Electronic Supplementary Material 2

Details of phylogeny construction

For character mapping, we constructed the composite phylogeny based on Miller-Butterworth *et al.* (2007) for inter-familial relationships. Where > 2 species from a family were present in our data set, we used Stoffberg *et al.* (2010) for species relationships within Rhinolophidae, Hoofer & Van Den Bussche (2003) and Stadelmann *et al.* (2007) for Vespertilionidae, and Jones *et al.* (2002) for all other families. We set branch lengths as equal because different methods were used to construct each of these phylogenies and because this is the preferred and most robust setting in CAIC when using incomplete phylogenies (see Purvis & Rambault 1995 and references therein).

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